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MDPI

Twelve-Year Landsat-Derived Surface Urban Heat Island Intensity by Local Climate Zone in Barcelona (2013–2024)

A 12-year, 30 m thermal baseline for urban climate adaptation

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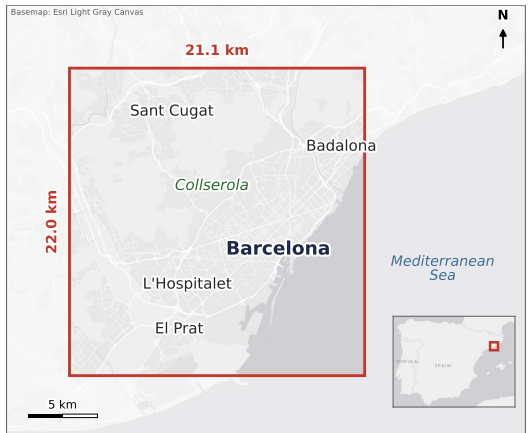
ECRS 2026 | Session S5 – Remote Sensing in Urban Areas | Oral, online

The Problem: No Baseline, No Way to Measure Change

- Outdoor thermal exposure depends strongly on urban surface temperature.
- The gap: long-term, high-resolution LST climatologies remain scarce for Mediterranean cities.
- Without a baseline: cannot measure change, cannot prioritise interventions, cannot tell which neighbourhoods are persistently hotter.
- Barcelona: 3M+ people in its metropolitan area, a Mediterranean coastal city – until now, no systematic, satellite-based thermal baseline.

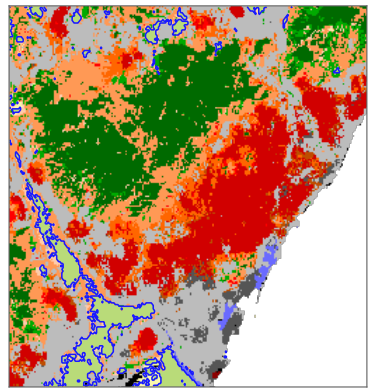
Study Area and Data

- **464 km^2** over the Barcelona metropolitan core (21.1 × 22.0 km).
- **378** Landsat 8/9 scenes, Collection 2 Level-2, 2013–2024 (12 years).
- 30 m grid, cloud-filtered.
- Four seasonal medians (winter, spring, summer, autumn) → one climatology.



Local Climate Zones + Two Consistency Checks

- Local Climate Zones (WUDAPT, Demuzere et al.): building height, density, vegetation cover.
- Reference: **LCZ D** (low plants) – the rural background.
- Two consistency checks:
  - ERA5-Land skin temperature, $n = 325$ pairs – seasonal cycle.
  - ECOSTRESS 70 m, $n = 7$ near-coincident scene pairs ($|\Delta t| \leq 2$ h).
- **Consistency – not validation.** Seven scene pairs are not enough for full validation.



RESULTS

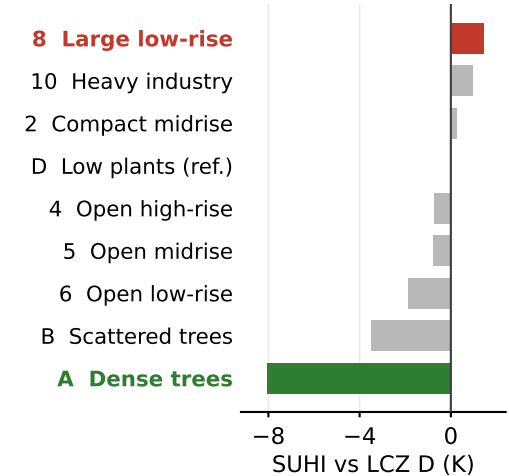
LCZ Thermal Hierarchy: The Tree-Canopy Cool Island

Summer LST vs. low plants (LCZ D)

+1.4 K
hottest built class (LCZ 8)

−8.0 K
dense trees (LCZ A)

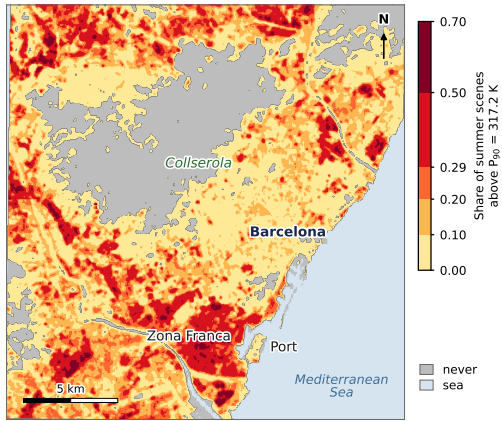
Trees cool far more than buildings heat.



Classes < 2 % of area omitted (C, F, E, 3, 1, 9).

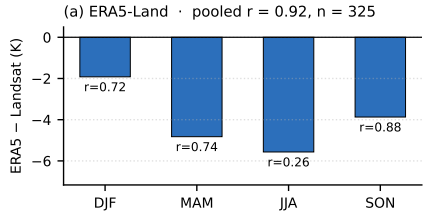
Hotspot Analysis: Where Heat Recurs

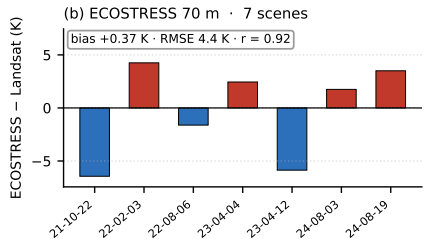
- Hotspot = pixel above the summer (JJA) P90 threshold.
- **78%** of the land area (64% of the window, which includes sea): at least one hotspot event in 12 years.
- **1.3%** of the land area persistently affected (hotspot frequency > 0.5) – the same pixels, year after year.
- Where adaptation matters most: tree planting, cool roofs, urban redesign.



Two Independent Consistency Checks

- **ERA5-Land (a):** pooled seasonal cycle $r = 0.92$.
  - Summer (JJA) $r = 0.26$: tracks the seasons, not the intra-urban structure (0.1° grid).
- **ECOSTRESS 70 m (b):** bias $+0.37$ K, RMSE 4.4 K, $r = 0.92$ – consistent with the 30 m field.
- ERA5 tells you the season. ECOSTRESS tells you the LST is about right. Only Landsat gives the 30 m structure.

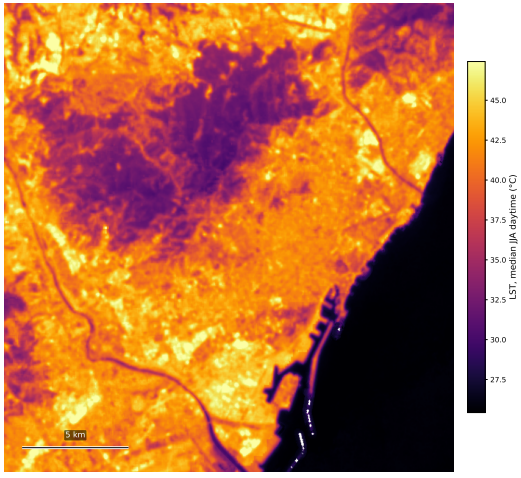




This is the baseline.

12 years. 30 meters.

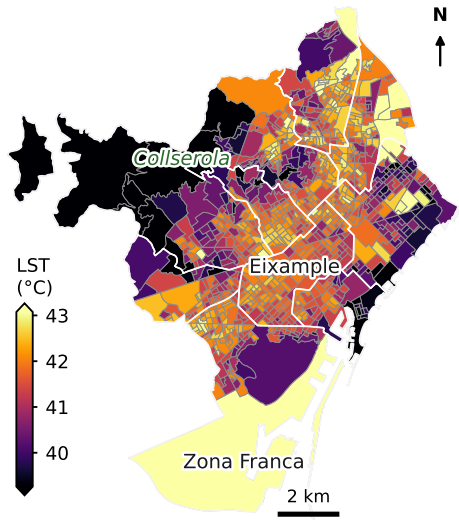
Median summer (JJA) daytime LST, 2013–2024.



Building Exposure: From Map to Baseline

- The climatology is not just a map – it is an exposure baseline.
- **106,000+** buildings across 13 municipalities; each gets the median summer LST in a 30 m buffer.
- At Landsat's effective thermal resolution (∼100 m): exposure indicators, not per-building retrievals.
- Which neighbourhoods carry the highest thermal load on their building stock.

Map: aggregation to census sections, Barcelona city ($n = 1{,}068$).



Three Findings – and One Takeaway

1. A 12-year, 30 m SUHI-by-LCZ climatology for a Mediterranean city – reproducible, built entirely from open data.
2. Dense tree cover is 8 K cooler than the rural reference ($\approx$9.4 K below large low-rise) – vegetation is the dominant spatial cooling signal.
3. ECOSTRESS confirms our Landsat field is consistent; ERA5 cannot substitute for high-resolution urban LST.

TAKEAWAY

A reproducible, satellite-based thermal baseline for adaptation planning in Mediterranean coastal cities.

Thank You

Slides, posters and figure code:

github.com/kerincardona/ecrs2026-barcelona-lst

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Happy to take your questions.